

VISHVANATH MADASAMY

Cloud Platform Software Engineer | Full-Stack Development | Containerization & Cloud Infrastructure
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PROFESSIONAL SUMMARY

Full-stack Software Engineer with hands-on experience building containerized cloud applications, RESTful backend services, and dynamic front-end interfaces. Proficient in JavaScript, Python, and SQL/NoSQL databases, with practical expertise in Docker, Kubernetes, and CI/CD pipeline management. Experienced in microservices architecture, system design, and agile development workflows. Strong communicator with a track record of mentoring peers and collaborating cross-functionally to deliver scalable, production-grade software. AWS Certified AI Practitioner & Cloud Practitioner. M.S. Data Science (GPA 3.778/4.0) — OPT-eligible May 2026.

TECHNICAL SKILLS

Languages: JavaScript (Node.js), Python, SQL, Java, C, HTML/CSS

Frontend & Frameworks: ReactJS, Angular (familiar), Next.js, d3.js — component-based UI, dynamic healthcare application development

Backend & APIs: Node.js, FastAPI, Flask, REST APIs, Microservices architecture, Authentication/Authorization, API integration

Containerization & Orchestration: Docker, Kubernetes — containerized application deployment, container orchestration, scalable cloud infrastructure

Databases (SQL & NoSQL): MySQL, PostgreSQL, MongoDB, OpenSearch, SQL Server (SSMS), Amazon Redshift — schema design, query optimization, data modeling

Cloud Platforms: AWS (EC2, S3, Lambda, Glue, Athena, QuickSight, SageMaker), Azure (Data Factory, Databricks), GCP

CI/CD & DevOps: Git, GitHub, CI/CD pipelines, Ansible, Chef, automated testing workflows — modern version control & agile delivery

System Design & Algorithms: Distributed systems, data structures & algorithms, scalable architecture patterns, security & data flow analysis

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-01) — Amazon Web Services | Active: Mar 2026 – Mar 2029

AWS Certified Cloud Practitioner (CLF-02) — Amazon Web Services | Active: Mar 2026 – Mar 2029

Microsoft Azure Fundamentals (AZ-900) — In Progress

PROFESSIONAL EXPERIENCE

Information Systems Development & Integration Assistant (*IT Systems Analyst*)— Cloud Platform & Systems Integration

| University of Massachusetts Dartmouth
North Dartmouth, MA | Sep 2025 – Present

- Developed and deployed containerized backend services using Docker and Kubernetes — managing CI/CD pipelines, Ansible/Chef configuration management, and automated testing to ensure reliable, scalable cloud platform operations across a 10,000+ user environment.
- Designed and implemented RESTful backend APIs and microservices in Python integrating enterprise systems (INFOR POS, UMass Pass, Lenel OnGuard) — engineering robust authentication/authorization flows and data security controls aligned with system requirements.
- Engineered and optimized SQL and NoSQL database systems (MySQL, PostgreSQL, MongoDB, OpenSearch) — designing schemas, tuning queries, and building efficient data access layers to support real-time platform functionality.
- Built ReactJS-based dynamic web application components with interactive drill-down visualizations and d3.js charting — delivering clear, high-quality interfaces enabling self-service analytics for non-technical stakeholders.
- Developed an AI-powered Log Analyzer (Python + PowerShell) leveraging LLM orchestration (Amazon Bedrock, LangChain) — implementing anomaly detection algorithms, geolocation threat visualization, and automated security alerting.
- Collaborated with architects to define cloud platform roadmap; mentored junior developers on software engineering best practices, version control workflows, and agile delivery — driving adoption of modern tooling across the team.

Data Analysts Intern | Value IBMS Pvt Ltd.

Chennai, India | Oct 2023 – Dec 2023

- Architected and deployed a production-grade backend ETL pipeline using Python and SQL Server — automating data ingestion, transformation, and loading workflows on cloud infrastructure, reducing data processing time by 45%.
- Built microservice-based REST API integrations with SQL Server backend — enabling real-time reporting pipelines and self-service analytics that contributed directly to a 30% increase in product sales outcomes.
- Designed and developed KPI dashboards with ML-based trend forecasting and automated reporting — writing efficient, clean code that translated complex data requirements into actionable business intelligence tools.
- Applied Python (Pandas, NumPy) with machine learning models for data cleaning, preprocessing, and predictive analysis — demonstrating capability to develop software solutions by studying data flows, system requirements, and security needs.

IT System Analyst Intern | Value IBMS Pvt Ltd.

Chennai, India | Jul 2023 – Aug 2023

- Deployed a serverless ML-based monitoring and automation solution for Pune Metro CPM infrastructure — achieving a 37% improvement in building management system efficiency through intelligent data-driven alerting.
- Built Python automation scripts for real-time IoT data ingestion, SQL query optimization, and automated report generation — eliminating manual monitoring processes and demonstrating strong problem-solving and independent development skills.

KEY PROJECTS

AI-Powered Medical Image Analysis Platform | Node.js, Python, Next.js, ReactJS, GCP, REST APIs Oct 2025

- Built a full-stack cloud-hosted application with a Next.js/ReactJS frontend, Python REST API backend, and LLM-powered inference pipeline on GCP — applying system design principles to architect secure, high-quality software for healthcare data analysis. [GitHub](#)

Boston Library Checkout Trends Analytics | Node.js, Python, SQL, AWS S3, Glue, Athena, Redshift, Lambda, QuickSight Aug 2025

- Engineered a fully serverless AWS cloud platform pipeline — ingesting 53 MB of CSV data into S3, orchestrating Spark ETL jobs via AWS Glue to produce partitioned Parquet tables, querying via Athena, and delivering QuickSight dashboards that visualized peak checkout hours and usage patterns.
- Designed SQL data models and transformations in Python — normalizing schemas, parsing timestamps, and deduplicating records to produce clean, query-optimized datasets; demonstrated proficiency with both SQL (Athena/Redshift) and NoSQL data paradigms.

Car Rental Management System | JavaScript, ReactJS, Python, MySQL, SQL Server May 2025

- Designed and developed a full-stack web application with JavaScript/ReactJS frontend and Python backend connected to MySQL/SSMS — implementing CRUD operations, dynamic pricing logic, real-time availability querying, and an admin analytics dashboard. [Github](#)

Library Assistant Chatbot — Conversational AI | Python, Node.js, Flask, REST API, Rasa NLU Jul 2024

- Designed and deployed an NLP-powered backend service using Rasa NLU and Flask REST APIs — implementing intent classification, entity recognition, and dialogue management with webhook integrations for fully automated self-service workflows. [Github](#)

EDUCATION

Master of Science in Data Science

Aug 2024 – May 2026

University of Massachusetts Dartmouth — North Dartmouth, MA | GPA: 3.778 / 4.0

Coursework: AI, Advanced Machine Learning, Database Design, Network Security, High Performance Scientific Computation, Data Structures & Algorithms, Statistical Analysis, Data Visualization

Bachelor of Engineering — Electronics & Communication Engineering

Jul 2020 – Apr 2024

Anna University — Chennai, India | GPA: 8.65 / 10 | Best Undergraduate Final Project Award 2024 (ML-based automated image detection system)